Skills Summary

Slope, Parallel Lines and Perpendicular Lines

Use this reference while completing your worksheet or when studying.

Skill 1 — Slope from two points

Rule. $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ — rise over run. Take both differences in the *same* order; swapping one of them flips the sign. A run of zero (two points with the same x) means a vertical line, whose slope is undefined.

Example. Find the slope of the line through $(1, 2)$ and $(5, 14)$.
Step 1. Want a slope from two points — subtract the y-values for the rise and the x-values for the run, keeping the same point first in both.
Step 2. $m = \dfrac{14-2}{5-1} = \dfrac{12}{4}$
$m = \mathbf{3}$

Try it. Find the slope of the line through $(-2, 6)$ and $(4, -3)$.

check: $m = -\frac{3}{2}$

Skill 2 — The parallel test

Rule. Two lines are parallel exactly when their slopes are *equal*. So a line parallel to a given one has the same slope — find the given line's slope and you are done; the points it passes through do not change it.

Example. Find the slope of any line parallel to the line through $(0, 1)$ and $(4, 7)$.
Step 1. Parallel means equal slopes, so the whole job is the slope of the line I was given.
Step 2. $m = \dfrac{7-1}{4-0} = \dfrac{6}{4}$
$m = \mathbf{\frac{3}{2}}$

Try it. Find the slope of any line parallel to the line through $(-1, 5)$ and $(3, -3)$.

check: $m = -2$

Skill 3 — The perpendicular test

Rule. Two lines are perpendicular exactly when $m_1 \times m_2 = -1$, that is, when each slope is the *negative reciprocal* of the other: flip the fraction and change the sign. Read straight off two points, the perpendicular slope is

$$-\frac{x_2 - x_1}{y_2 - y_1}.$$

Example. Find the slope of any line perpendicular to the line through $(2, 1)$ and $(6, 9)$.
Step 1. Perpendicular means negative reciprocal, so I can swap rise and run and change the sign in one step instead of computing the slope first.
Step 2. $m = -\dfrac{6-2}{9-1} = -\dfrac{4}{8}$

$m = \boldsymbol{-\frac{1}{2}}$

Try it. Find the slope of any line perpendicular to the line through $(-4, 3)$ and $(2, -3)$.

check: $m = \mathbf{1}$

(!) Watch out: the negative reciprocal needs *both* moves. Flipping $\frac{2}{3}$ to $\frac{3}{2}$ without the minus sign, or negating to $-\frac{2}{3}$ without flipping, gives a line that is not perpendicular.

Skill 4 — Classifying a figure by slopes

How the tests classify a quadrilateral. Compute the slope of each side, then read off: both pairs of opposite sides with equal slopes means a parallelogram; exactly one pair means a trapezoid; a parallelogram whose adjacent slopes multiply to -1 is a rectangle.

Example. In quadrilateral $ABCD$ the slope of $\overline{DC}$ is $\frac{1}{3}$, and A is at $(0, 1)$ with B at $(6, 3)$. Find the slope of $\overline{AB}$, and say whether $\overline{AB} \parallel \overline{DC}$.
Step 1. $\overline{AB}$ and $\overline{DC}$ are opposite sides, so the parallel test applies: compute the slope of $\overline{AB}$ and compare.
Step 2. $m_{AB} = \dfrac{3-1}{6-0} = \dfrac{2}{6}$, which equals the slope of $\overline{DC}$, so the two sides are parallel.

$m_{AB} = \boldsymbol{\frac{1}{3}}$

Try it. In quadrilateral $ABCD$ the slope of $\overline{DC}$ is $\frac{1}{2}$, and A is at $(-2, 0)$ with B at $(4, 3)$. Find the slope of $\overline{AB}$.

check: $m_{AB} = \boldsymbol{\frac{1}{2}}$